WILP, BITS Pilani

MACHINE LEARNING · COMPANION READER

Decision Tree Classifier & Information Theory

Hunt's Algorithm, Entropy, Information Gain, and ID3 Construction

Session 7 · Read alongside the lecture slides · Every slide example worked out in full

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

1 First Taste: Twenty Questions for Data

A decision tree splits complex decisions into a sequence of simple yes/no questions.

A **decision tree** converts tabular data into a flowchart structure. Each internal node tests an attribute, each branch represents an outcome, and each leaf node holds a class prediction.

☞ The intuition

A decision tree breaks down a complex classification problem into a hierarchy of simple feature tests.

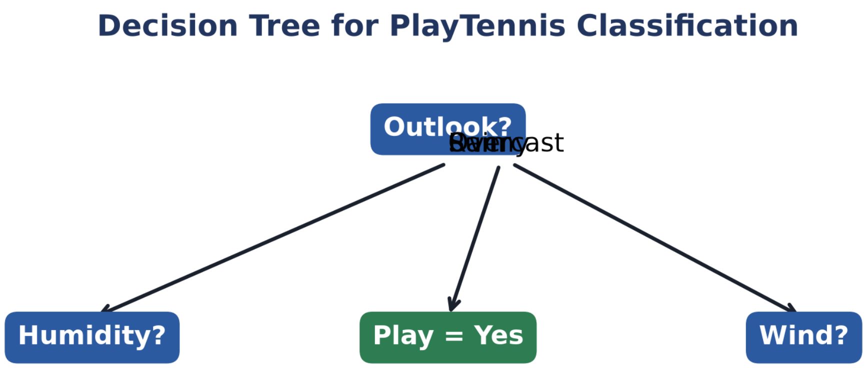

Figure 1: Structure of a decision tree showing decision nodes, feature branches, and pure leaf node predictions.

2 Decision Tree Representation & Expressiveness

Decision trees express disjunctions of conjunctions on attribute constraints.

Any decision tree can be rewritten as a set of logical rules. Each path from root to leaf forms a conjunction (AND) of attribute tests.

2.1 Hunt's Algorithm

Most decision tree algorithms are variations of **Hunt's Algorithm**. Given dataset D_t at node t:

1. If all records in D_t belong to class y_k, node t becomes a leaf labeled y_k.
2. If D_t contains multiple classes, select an attribute test condition to partition D_t into child subsets and recurse.

3 Information Theory & Entropy

Entropy measures the amount of impurity, disorder, or unexpected surprise in a dataset.

For a dataset S with c classes where p_i is the proportion of class i:

$$H(S) = -\sum_{i=1}^{c} p_i \log_2(p_i). \quad (1)$$

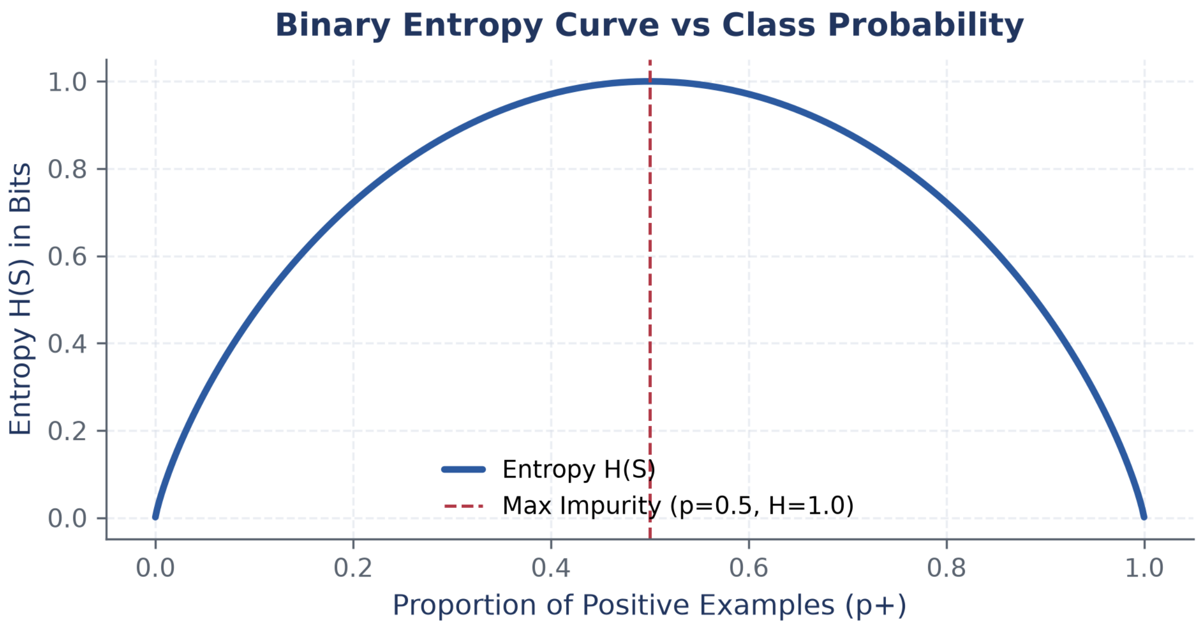

Figure 2: Binary entropy curve H(S) as a function of positive class proportion p+. Impurity peaks at p+=0.5.

✎ Worked example: Computing Entropy for Sample Class Distributions

Calculate binary entropy for three representative subsets:

Step 1. Pure dataset $[14+, 0-]$**:** $H(S) = 0$ bits.

Step 2. Balanced dataset $[7+, 7-]$**:** $H(S) = 1.0$ bit.

Step 3. Slightly imbalanced dataset $[9+, 5-]$**:** $H(S) \approx 0.940$ bits.

4 Information Gain & The ID3 Algorithm

Information Gain measures the expected reduction in entropy from splitting on an attribute.

The **ID3** algorithm selects the attribute that maximizes **Information Gain**:

$$\mathrm{Gain}(S, A) = H(S) - \sum_{v \in \mathrm{Values}(A)} \frac{|S_v|}{|S|} H(S_v). \tag{2}$$

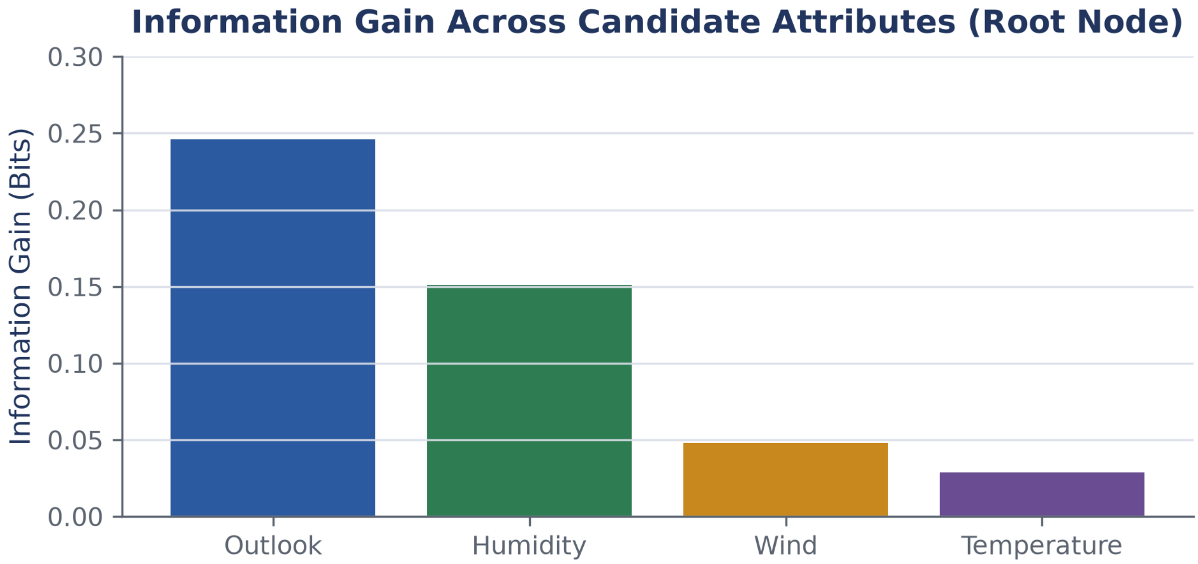

Figure 3: Information Gain values for candidate attributes at the root node. Outlook yields the highest gain.

✎ **Worked example: 14-Sample PlayTennis Root Split Selection**

In the 14-sample PlayTennis dataset ($S = [9+, 5-]$, $H(S) = 0.940$ bits), evaluate root split candidates:

Step 1. Evaluate Outlook: $\mathrm{Gain}(S, \mathrm{Outlook}) = 0.940 - 0.694 = 0.246$ bits.

Step 2. Compare Root Candidates: Outlook (0.246), Humidity (0.151), Wind (0.048), Temp (0.029). Outlook is chosen as root.

✓ **Key takeaway**

ID3 recursively chooses the attribute with maximum Information Gain at each node.

Glossary & Symbols

Key Terms

Decision Tree A non-parametric supervised model representing decisions as a hierarchical tree structure.

Entropy A statistical measure of dataset impurity.

Information Gain The expected reduction in entropy resulting from partitioning examples along an attribute.